compiled by Ahmed Gul www.RocketRevise.com

Questions

Q1.

Prove by contradiction that, if a, b are positive real numbers, then $a + b \geqslant 2\sqrt{ab}$

(Total for question = 4 marks)

RocketRevise

(**Q06 WMA14/01, Specimen papers**)

Q2.

Given that n is an integer, use algebra, to prove by contradiction, that if n^3 is even then n is even.

(Total for question = 4 marks)

(**Q01 WMA14/01, Oct 2020**)

Q3.

www.RocketRevise.com

(i) Relative to a fixed origin O, the points A, B and C have position vectors **a**, **b** and **c** respectively.

Points A, B and C lie in a straight line, with B lying between A and C.

Given $AB : AC = 1 : 3$ show that

$$\mathbf{c} = 3\mathbf{b} - 2\mathbf{a}$$

(3)

(ii) Given that $n \in \mathbb{N}$, prove by contradiction that if n^2 is a multiple of 3 then n is a multiple of 3

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(5)

(Total for question = 8 marks)

(Q09 WMA14/01, June 2021)

Q4.

RocketRevise

Prove by contradiction that there is no greatest odd integer.

(2)

(Total for question = 2 marks)

(Q03 WMA14/01, Jan 2021)

Q5.

(a) A student's attempt to answer the question

"Prove by contradiction that if n^3 is even, then n is even"

is shown below. Line 5 of the proof is missing.

Assume that there exists a number n such that n^3 is even, but n is odd.

If n is odd then $n = 2p + 1$ where $p \in \mathbb{Z}$

So $n^3 = (2p + 1)^3$

$= 8p^3 + 12p^2 + 6p + 1$

=

This contradicts our initial assumption, so if n^3 is even, then n is even.

www.RocketRevise.com

Complete this proof by filling in line 5.

www.RocketRevise.com

(1)

(b) Hence, prove by contradiction that $\sqrt[3]{2}$ is irrational.

(5)

(Total for question = 6 marks)

(**Q10 WMA14/01, Oct 2021**)

RocketRevise

Q6.

Three **consecutive** terms in a sequence of real numbers are given by

$$k, \quad 1 + 2k \quad \text{and} \quad 3 + 3k$$

where k is a constant.

Use proof by contradiction to show that this sequence is not a geometric sequence.

(Total for question = 5 marks)

(**Q06 WMA14/01, Jan 2022**)

Q7.

www.RocketRevise.com

A student was asked to prove by contradiction that

"there are no positive integers x and y such that $3x^2 + 2xy - y^2 = 25$"

The start of the student's proof is shown in the space below.

compiled by Ahmed Gul www.RocketRevise.com

Assume that integers x and y exist such that $3x^2 + 2xy - y^2 = 25$

$$\Rightarrow (3x - y)(x + y) = 25$$

If $(3x - y) = 1$ and $(x + y) = 25$

$$\left.\begin{matrix} 3x - y = 1 \\ x + y = 25 \end{matrix}\right\} \Rightarrow 4x = 26 \Rightarrow x = 6.5, y = 18.5 \quad \text{Not integers}$$

Show the calculations and statements that are needed to complete the proof.

(4)

RocketRevise

(Total for question = 4 marks)

(Q08 WMA14/01, Oct 2022)

Q8.

A student was asked to prove, for $p \in \mathbb{N}$, that

"if p^3 is a multiple of 3, then p must be a multiple of 3"

The start of the student's proof by contradiction is shown in the space below.

Assumption:

There exists a number p, $p \in \mathbb{N}$, such that p^3 is a multiple of 3, and p is NOT a multiple of 3

Let $p = 3k + 1$, $k \in \mathbb{N}$.

www.RocketRevise.com

Consider $p^3 = (3k + 1)^3 = 27k^3 + 27k^2 + 9k + 1$

$= 3(9k^3 + 9k^2 + 3k) + 1$ which is not a multiple of 3

(a) Show the calculations and statements that are required to complete the proof.

(3)

(b) Hence prove, by contradiction, that $\sqrt[3]{3}$ is an irrational number.

(5)

www.RocketRevise.com

(Total for question = 8 marks)

(Q09 WMA14/01, Jan 2023)

Q9.

Use proof by contradiction to prove that $\sqrt{7}$ is irrational.

RocketRevise

(You may assume that if k is an integer and k^2 is a multiple of 7 then k is a multiple of 7)

(4)

(Total for question = 4 marks)

(Q07 WMA14/01, June 2023)

Q10.

With respect to a fixed origin O, the equations of lines l_1 and l_2 are given by

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 8 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -4 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 4 \\ 8 \end{pmatrix}$$

www.RocketRevise.com

where λ and μ are scalar parameters.

Prove that lines l_1 and l_2 are skew.

(5)

(Total for question = 5 marks)

(**Q09 WMA14/01, Oct 2022**)

Q11.

With respect to a fixed origin O, the lines l_1 and l_2 are given by the equations

$$l_1: \quad \mathbf{r} = \begin{pmatrix} -1 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} \qquad l_2: \quad \mathbf{r} = \begin{pmatrix} 2 \\ -2 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -3 \\ b \end{pmatrix}$$

RocketRevise

where λ and μ are scalar parameters and b is a constant.

Prove that for all values of $b \neq 7$, the lines l_1 and l_2 are skew.

(6)

(Total for question = 6 marks)

(**Q08 WMA14/01, Jan 2021**)

Q12.

Use proof by contradiction to show that, when n is an integer,

$$n^2 - 2$$

is **never** divisible by 4

www.RocketRevise.com

(Total for question = 4 marks)

(**Q09 WMA14/01, June 2022**)

compiled by Ahmed Gul www.RocketRevise.com

Q13.

(a) Prove by contradiction that for all positive numbers k

$$k + \frac{9}{k} \geqslant 6$$

(4)

(b) Show that the result in part (a) is not true for all real numbers.

(1)

(Total for question = 5 marks)

(Q04 WMA14/01, Oct 2023)

www.RocketRevise.com

www.RocketRevise.com